

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-09

BOTANY

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Topics To Covered

- 1** Virus, Viroids, Prions and Lichens ✓

INTRODUCTION

≡ Virus, viroids, prions and lichens — did not get any place in 5-Kingdom Classification proposed by R.H Whittaker.

VIRUSES

DNA → RNA → protein → Enzyme

Ribosome
required

"-nt in viruses"

≡ Study of virus = virology.

≡ These are infectious agents — when +nt inside host.

≡ Viruses — acellular/noncellular — Thus exception to cell theory.

≡ Viruses — Enter the host cell and control the machinery of host cell.

≡ Viruses — Obligate parasite (i.e., completely dependent on host cell).

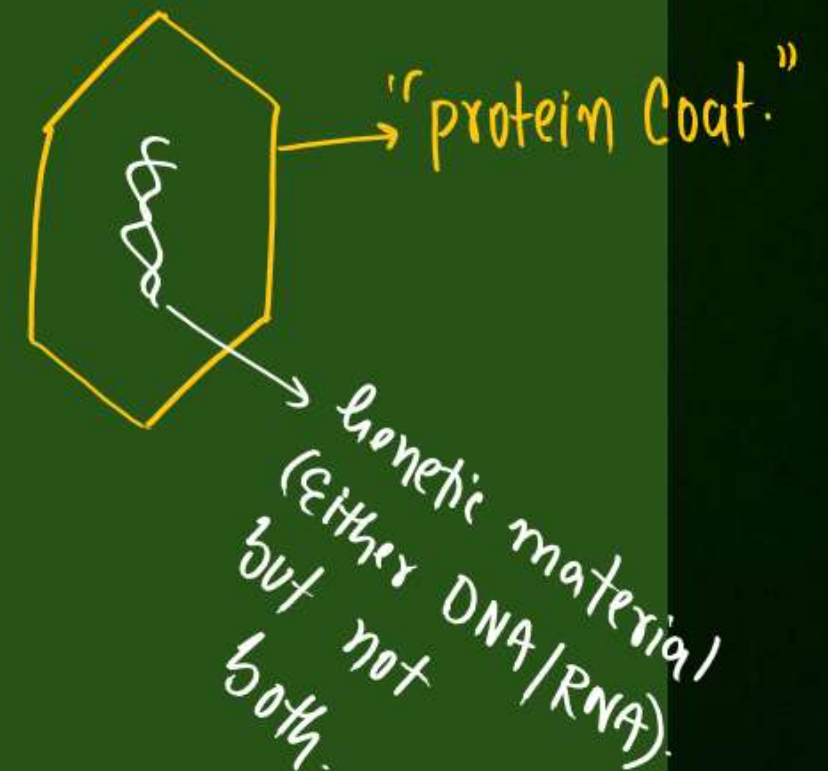
≡ Viruses — intracellular parasite (i.e., inside host cell).

≡ Viruses — Contain protein coat and genetic material.

≡ Viruses — • living inside host cell.

but

• dead (inert) outside host cell.



Q. Would you call viruses — living or non-living?

Ans. Viruses — Considered as **borderline** b/w livings and non-livings because they show characters of both.

- **living** inside host cell.
- **non-living** outside host cell.

* Virus — means **venom** / **poisonous fluid**.

DISCOVERY OF VIRUS

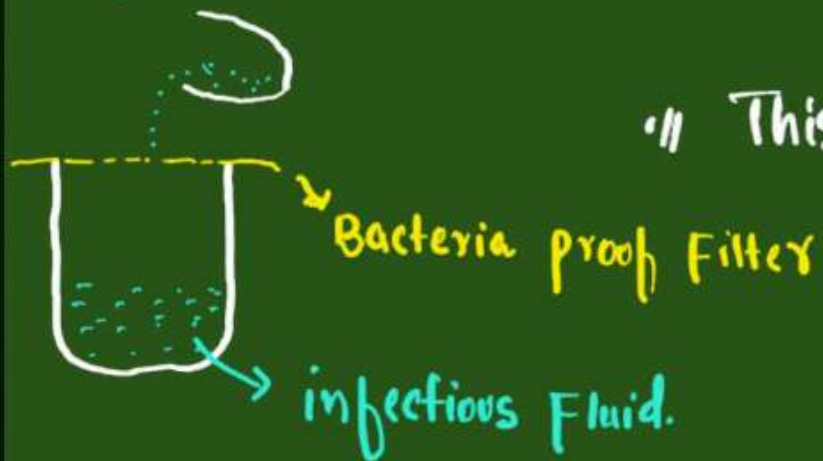
! Dmitri Ivanowsky : (1892)

• He observed mosaic disease in Tobacco leaf (caused by certain microbes).



• He isolated chemical (extract) from infectious leaf and then passed it through bacterial filter paper.

• He concluded — these microbes were smaller than bacteria because they pass through the bacteria proof-filter.



• This microbe later discovered to be TMV (Tobacco Mosaic Virus).

↓
"First discovered virus"

②. M.W. Beijerinck: (1898)

↓ ↳ Father of **Virology**.

demonstrated that — Extract of infected leaves of tobacco plant Causes
fresh infection in healthy plants.

• He used the term **"virus"** for this pathogen. (disease causing organism).

• He called infectious fluid as **"Contagium Vivum fluidum"**

↓ means

infectious living fluid.
(living-only when ~~not~~ inside host)...

③. W.M Stanley :- (1935)

- He first crystallised viruses.
- These virus crystals consist

of

largely proteins.

and

Nucleic Acid.

→ DNA / RNA

STRUCTURE OF VIRUS

∥ Virus = "Nucleoprotein" (protein + Nucleic acid)



STRUCTURE

1. Protein Coat (External covering)

∴ AKA Capsid

∴ Capsid - made of small subunits kld as Capsomers.

∴ Arrangement of Capsomers:

✓ a. Helical ex. TMV

✓ b. Helical + polyhedral ex. Bacteriophage

NOTE: No. of capsomers is characteristic feature of viruses.

2. Genetic material

∥ Genetic material is infectious.. ✓

∥ Either DNA | RNA (but not both).

DNA virus

SSDNA virus

ex. φx174

dsDNA virus

ex. Bacteriophage

RNA virus

SSRNA virus (max.)

ex. HIV ✓

Influenza ✓

TMV ✓

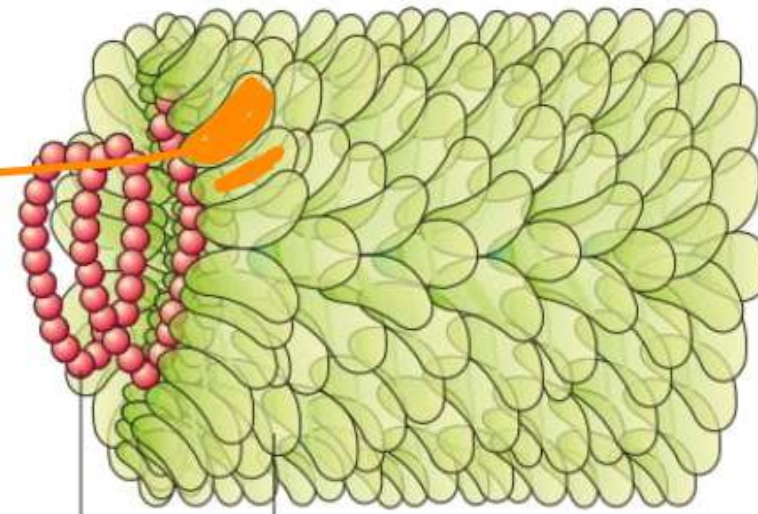
DSRNA virus

ex. Rotaviruses. ✓

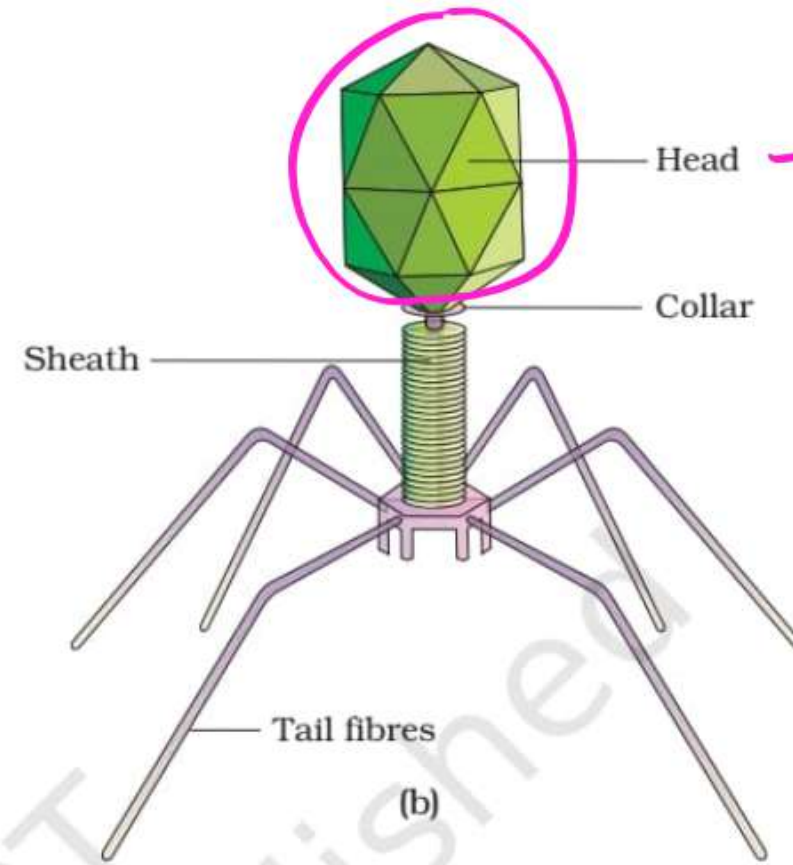
∴ plant viruses — generally ssRNA.

∴ Animal viruses — ssRNA, dsRNA, dsDNA...

Capsomer



RNA
(SSRNA) **TMV**
(a)



Head → Polyhedral

Figure 2.6 (a) Tobacco Mosaic Virus (TMV) (b) Bacteriophage

Bacteriophage
↓
Virus which
infects bacteria

* Viral diseases:

Mumps, Small pox, Herpes, Influenza, AIDS (in humans).

* Viral Symptoms (plants):

- Mosaic formation (patches)
- Rolling & Curling of leaves. ✓
- Yellowing of leaves (chlorosis) ✓
- Vein clearing. ✓
- Stunted growth and dwarfing... ✓

SUB-VIRAL AGENTS

a. Viroids:

- Discovered by T.O Diener (1971).
- Infectious particle not protein coat.
- Infectious particle is RNA.
- low mol. weight.
- Smaller than viruses.
- Causes "PSTD" - Potato spindle tuber disease.

b. Prions:

- Infectious protein particles not genetic material.
- Size equal to viruses.
- Causes infection by changing the folding of other proteins.
- Causes neurological disorder in cattle - BSE
 ↓
 Bovine Spongiform Encephalopathy
 ↓
 AKA Mad cow disease.
- Causes neurological disorder in humans - CJD
 ↓
 Creutzfeldt Jacob Disease.
 ↓
 Analogous to BSE..

LICHENS

- ≡ Symbiotic relationship b/w Algae and fungi.
- ≡ Mutualism (+, +).
- ≡ Here both partners (algae & Fungi) — get benefitted.
- ≡ Algal partner (phycobiont) — performs photosynthesis
- ≡ Fungal partner (Mycobiont) — obtains H_2O and minerals from Soil.
provides Shelter.
- ≡ Lichens — are good pollution indicators (they do not grow in polluted areas)...

QUESTIONS

^{BSE}
Mad cow disease in cattle and Cr Jacob disease in humans are due to infection by _____. [2022 Re]

- ☒ 1 Prion
- ☐ 2 Bacterium
- ☐ 3 Virus
- ☐ 4 Viroid

QUESTIONS

Which of the following is correct about viroids?

[MR* 2020]

- ✓ 1 They have free RNA without protein coat
- 2 They have DNA with protein coat ✗
- 3 They have free DNA without protein coat ✗
- 4 They have RNA with protein coat. ✗

QUESTIONS

H/w



Which one of the following statements about viruses is correct?

[2023]

- 1 Viruses possess their own metabolic system
- 2 All viruses contain both RNA and DNA
- 3 Viruses are obligate parasites
- 4 Nucleic acid of viruses is known as capsid

QUESTIONS

Hlw



Virus are living because:

[2000]

- 1 They multiply in host cells
- 2 Carry anaerobic respiration
- 3 Carry metabolic activity
- 4 Cause infection

QUESTIONS

H/W



Which disease of man is similar with cattle's, bovine spongiform encephalopathy?
[OS 2000]

- 1 Encephalitis
- 2 Jacob-creutzfeldt disease
- 3 Spongiocitis of cerebrum
- 4 Spondylitis

QUESTIONS

H/w



Lichens can be used as:

[1999]

- 1 Bio-indicator for water and air pollution
- 2 Initial vegetation for waste lands
- 3 Source of wood
- 4 To check the air pollution

QUESTIONS

A/w



Tobacco mosaic virus (TMV) genes are:

[1994]

- 1 Single stranded RNA
- 2 Double stranded DNA
- 3 Proteinaceous
- 4 Double stranded RNA

QUESTIONS

Hlw



Which of the following statement is incorrect?

[2019]

- 1 Viroids lack a protein coat
- 2 Viruses are obligate parasites
- 3 Infective constituent in viruses is the protein coat
- 4 Prions consist of abnormally folded proteins.

HOMEWORK



6 Questions

Thank
You